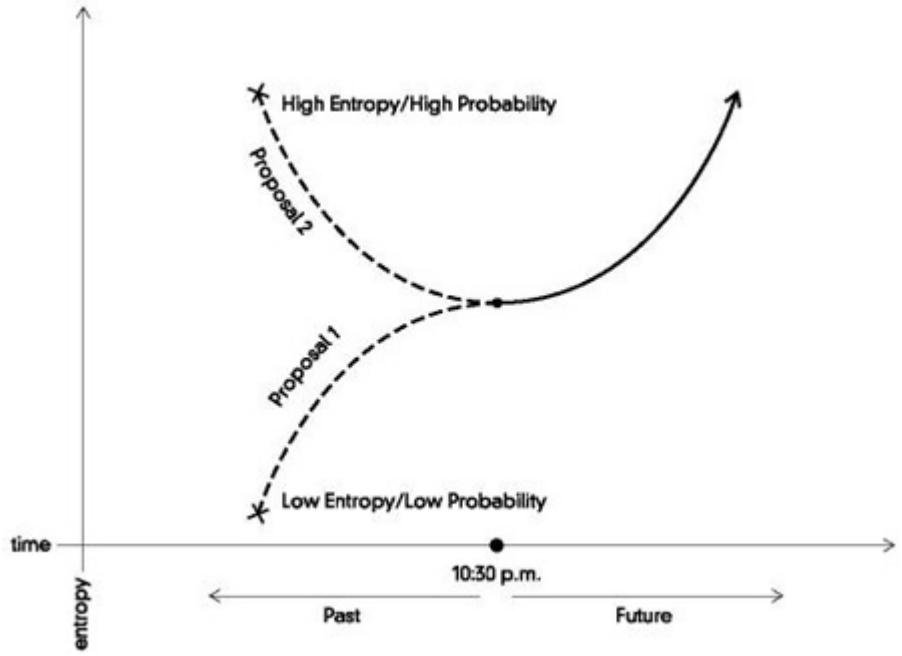


Figure 6.3 A comparison of two proposals for how the ice cubes got to their partially melted state, right now, at 10:30 p.m. Proposal 1 aligns with your memories of melting ice, but requires a comparatively low-entropy starting point at 10:00 p.m. Proposal 2 challenges your memories by describing the partially melted ice you see at 10:30 p.m. as having coalesced out of a glass of water, but starts off in a high-entropy, highly probable configuration of disorder at 10:00 p.m. Every step of the way toward 10:30 p.m., Proposal 2 involves states that are more likely than those in Proposal 1—because, as you can see in the graph, they have higher entropy—and so Proposal 2 is statistically favored.

It was a small step for Boltzmann to realize that the whole of the universe is subject to this same analysis. When you look around the universe right now, what you see reflects a great deal of biological organization, chemical structure, and physical order. Although the universe could be a totally disorganized mess, it's not. Why is this? Where did the order come from? Well, just as with the ice cubes, from the standpoint of probability it is extremely unlikely that the universe we see evolved from an even more ordered—an even less likely—state in the distant past that has slowly unwound to its current form. Rather, because the cosmos has so many constituents, the scales of ordered versus disordered are magnified intensely. And so what's true at the bar is true with a vengeance for the whole universe: it is *far* more likely—breathtakingly more likely—that the whole universe we now see arose as